

Joan Alcaide-Nunez

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Last updated: April 2026

EDUCATION

Ludwig-Maximilians-Universitat (LMU) Munich

Sept 2025 – Present

B.Sc. in Physics

German School of Barcelona

Sept 2017 – May 2025

Dual High School Diploma (Germany + Spain), GPA: 1.0/1.0

Additional Theoretical Training:

- *Gravity and Black Holes* — Perimeter Institute (GoPhysics!), 2025
- *Mathematics, Statistics, and Data Science* — Autonomous University of Barcelona (UAB), 2022
- *Algorithms and Programming (C++)* - Polytechnic University of Catalonia (UPC), 2022
- League of Codes (C++) - Harbour Space, 2022-2023

RESEARCH EXPERIENCE

Theoretical Probes of Spacetime (Independent Study)

Proposed July – Sept 2026

Supervision: Independent Project

- **Objective:** Deriving Modified Dispersion Relations (MDR) for high-energy photons in non-standard cosmological backgrounds.
- **Formalism:** Applying Hamilton-Jacobi equations to determine cumulative time-of-flight delays (Δt) for gamma-rays under Quintessence-based FLRW metrics.
- **Innovation:** Investigating the analytical coupling between the dark energy equation of state (w) and Planck-scale energy-dependent velocities.

GRB Cosmography & Relativistic Foundations

June – July 2025 [cite: 8, 10, 87]

Osservatorio Astronomico di Brera (OAB-INAF), Italy [cite: 9]

- **Mentorship:** Received training on relativistic fireball effects and the theoretical foundations of Gamma-Ray Bursts (GRBs)[cite: 13, 89].
- **Parameterization:** Constrained cosmological density parameters (Ω_m, Ω_Λ) via χ^2 minimization of empirical energy relations[cite: 11, 87].
- **Synthesis:** Integrated GRB results with simulated populations from Einstein Telescope binary neutron star expectations[cite: 12, 88].
- *Supervisors: Dr. Giancarlo Ghirlanda, Dr. Om Sharan Salafia* [cite: 14]

CAPIBARA Collaboration (Founder & PI)

2024 – Present [cite: 15, 17, 90]

- Leading student collaboration in developing instrumentation and mission concepts for high-energy astronomy CubeSat missions[cite: 18, 19, 90].

Type Ia Supernovae: Mapping the Expansion History

June – July 2024 [cite: 25, 27, 93]

Institute of Space Sciences (ICE-CSIC-IEEC), Spain [cite: 26]

- **Metric Fitting:** Benchmarked the empirical consistency of the local expansion history using SNe Ia light-curve fitting from ESO, ATLAS, and ZTF[cite: 28, 93].
- **Formalism:** Explored statistical foundations of the Hubble-Lemaitre Law through standardizable candle analysis[cite: 29, 94].
- *Supervisor: Dr. Lluís Galbany* [cite: 30, 95]

Cepheid Variable Stars Research

Aug – Dec 2023 [cite: 31, 33, 95]

Youth & Science Program [cite: 32]

- **Analytical Foundations:** Reconstructed the cosmic distance ladder by deriving the Period-Luminosity relationship from NASA-IPAC NED data[cite: 34, 95].
- **Methodology:** First exposure to rigorous scientific writing and the peer-review process ($H_0 = 70.26 \pm 7.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$)[cite: 34, 35, 96].

Student Internship at German Space Agency (DLR)

Oct 2024 [cite: 20, 22, 91]

Institute for Remote Sensing (DLR-IMF), Germany [cite: 21]

- Worked with photogrammetry departments on hyperspectral imaging and LandSat thermal analysis[cite: 23, 91].

AWARDS & HONORS

Youth & Science Fellowship: 3-year funded fellowship for research stays internationally.

Abiturpreis German Physical Society (DPG): Physics Distinction for Academic Achievement in high school.

IAAC: Silver Honour and National Award for Spain (2025), top 7% of 12,000 students. Similar awards in 2023 and 2024.

Jugend Forscht: First Prize Iberia (2024/2025), First prizes in regional and national rounds, Special Award in Scientific Photography (NRW 2025).

TALKS & OUTREACH

SSEA Symposium: The CRD Instrument and the COSMOS program, April 2026.

CosmoXarxa: High-energy astrophysics presentation at Barcelona Science Museum, March 2025.

Invited Talk: Intro to Astrophysics for 11th graders, German School Barcelona, 2024.

Outreach: Space Exploration Activity for 3rd grades, Escola Canigó Public School, 2023.

Explainers (Volunteer): Science outreach at CosmoCaixa Museum, 2022–2023. Slides of most talks can be found on my website.

SKILLS

Programming: Python (*numpy*, *scipy*, *pandas*, *matplotlib*, *astropy*), C++, LaTeX, Markdown.

Tools: Git/GitHub, Visual Studio Code, NeoVim, Linux environment.